

An evaluation of approaches for modelling hydrological processes in high-elevation, glacierized Andean watersheds

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Abstract:

We use two hydrological models of varying complexity to study the Juncal River Basin in the Central Andes of Chile with the aim to understand the degree of conceptualization and the spatial structure that are needed to model present and future streamflows. We use a conceptual semi-distributed model based on elevation bands [Water Evaluation and Planning (WEAP)], frequently used for water management, and a physically oriented, fully distributed model [Topographic Kinematic Wave Approximation and Integration ETH Zurich (TOPKAPI-ETH)] developed for research purposes mainly. We evaluate the ability of the two models to reproduce the key hydrological processes in the basin with emphasis on snow accumulation and melt, streamflow and the relationships between internal processes. Both models are capable of reproducing observed runoff and the evolution of Moderate-resolution Imaging Spectroradiometer snow cover adequately. In spite of WEAP's simple and conceptual approach for modelling snowmelt and its lack of glacier representation and snow gravitational redistribution as well as a proper routing algorithm, this model can reproduce historical data with a similar goodness of fit as the more complex TOPKAPI-ETH. We show that the performance of both models can be improved by using measured precipitation gradients of higher temporal resolution. In contrast to the good performance of the conceptual model for the present climate, however, we demonstrate that the simplifications in WEAP lead to error compensation, which results in different predictions in simulated melt and runoff for a potentially warmer future climate. TOPKAPI-ETH, using a more physical representation of processes, depends less on calibration and thus is less subject to a compensation of errors through different model components. Our results show that data obtained locally in *ad hoc* short-term field campaigns are needed to complement data extrapolated from long-term records for simulating changes in the water cycle of high-elevation catchments but that these data can only be efficiently used by a model applying a spatially distributed physical representation of hydrological processes. Copyright © 2013 John Wiley & Sons, Ltd.

KEY WORDS physically oriented modelling; conceptual modelling; mountain hydrology; model evaluation; data scarcity

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INTRODUCTION

Hydrological processes in high-elevation watersheds are difficult to study because of data scarcity, accessibility difficulties and complex topography. Modelling is affected by additional uncertainty due to the complexity of the process spatial variations (Klemes, 1990; Gurtz, 1999; Weingartner *et al.*, 2007). However, hydrological models are a useful tool for the study of these watersheds as they provide insight into the main catchment dynamics and responses to climatic forcing; these insights are especially valuable in the context of a changing climate, where catchment hydrological response may differ from that inferred from historical records (Vicuña *et al.*, 2010; Schaefli *et al.*, 2011).

The scarcity of information about meteorological conditions, snowmelt and glacier melt processes, ground-water and the magnitude of evapotranspiration restricts the predictive skills and accuracy of distributed glacio-hydrological models (Sivapalan, 2003b; Winsemius *et al.*, 2009; Magnusson *et al.*, 2011). The application of conceptual models gives rise to problems of overparametrization, parameter uncertainty and model output uncertainty (Beven, 1993, 2001; Seibert and McDonnell, 2002; Magnusson *et al.*, 2011). It is often the case that modelling studies for decision-making and water resources planning in high-elevation watersheds make simplifying assumptions, such as ignoring the presence of glaciers (Kling *et al.*, 2006; Lehning *et al.*, 2006; Bartolini *et al.*, 2011) or the role of soils and groundwater (e.g. Gascoin *et al.*, 2011), because of the lack of historical data or a perceived excessive difficulty in obtaining new data for model construction and validation (Sivapalan, 2003a). For these reasons, there is

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no conclusive evidence as to what the best modelling approach is in data-scarce regions. Although the scientific community has produced a sizable amount of research dealing with the interplay of data availability, model uncertainty and water resources planning under uncertainty, the local nature of many hydrological problems calls for continued research in this topic.

In the snow-and-glacier-dominated Central Andes of Chile, very little is known about the spatio-temporal dynamics and the hydrological role of glaciers and snow (Masiokas *et al.*, 2006, 2010). In general, but more pressing during dry periods, a quantification of hydrological processes is of key relevance for the region (Escobar *et al.*, 2000; Masiokas *et al.*, 2006; Gascoin *et al.*, 2011), but meteorological records are scarce and estimations therefore affected by large uncertainties. For these reasons, several operational studies have made use of simplified approaches to quantify runoff from the high-elevation headwaters of the Andes (CEPAL, 2009; Vicuña *et al.*, 2010, 2012). However, under climate change scenarios, changes in temperature and precipitation may result in different future behaviours of high-elevation basins (Beniston, 2003; Barnett *et al.*, 2005). The region's dependence on water resources provided by snowmelt and ice melt during summer months makes it vulnerable to changes in the cryospheric components of the water balance (Rivera *et al.*, 2002; Carrasco *et al.*, 2005; Masiokas *et al.*, 2006; Pellicciotti *et al.*, 2007; Bown *et al.*, 2008).

The physically oriented, distributed model Topographic Kinematic Wave Approximation and Integration at ETH Zurich (TOPKAPI-ETH) was applied by Ragetti and Pellicciotti (2012) to simulate glacier and snowmelt contributions to streamflow in the Upper Aconcagua River Basin of Central Chile over two ablation seasons. The authors showed that information from short-term glacio-meteorological experiments within the region is indispensable for the estimation of parameter values in distributed, physically oriented glacio-hydrological models. Meanwhile, studies with application of conceptual (Melo *et al.*, 2010; Vicuña *et al.*, 2010, 2012) or physically oriented but semi-distributed (Stehr *et al.*, 2008, 2009, 2010) hydrological models are relatively abundant in the Central Andes of Chile. Because streamflow data are relatively available, but field measurements represent a considerable financial and logistical effort, such models have often been favoured over more complex distributed models (Sivapalan, 2003a), in spite of the risk of overparametrization resulting, for example, in the adequate fit between observed and modelled streamflow but overall in a less realistic simulation of catchment behaviour (Seibert and McDonnell, 2002).

In this work, we evaluate two approaches previously applied for hydrological modelling in high-elevation basins of the Central Andes, with the aim of understanding the degree of conceptualization and spatial structure needed

to model present and future streamflows in such catchments. We assess their capability to contribute to the understanding of relevant hydrological processes in the region and expected changes in the water cycle. A first approach includes the conceptual, semi-distributed model Water Evaluation and Planning (WEAP), a general-objective model frequently used in water resource management, calibrated against observations of snow cover (SC) and streamflow. The second approach makes use of the distributed and physically oriented model TOPKAPI-ETH, moderately calibrated and informed by data from literature as well as data obtained locally, in *ad hoc* short-term field campaigns. The two models differ in spatial structure, temporal resolution and degree of conceptualization of processes. In comparison with TOPKAPI-ETH, WEAP does not explicitly represent glaciers and glacier melt. The same version of the model has been used previously also for Central Andean watersheds with the justification that these processes are minor and the scarce knowledge about them does not enable adequate calibration of the related model parameters (Vicuña *et al.*, 2010; Cortés *et al.*, 2012).

The two approaches are evaluated with respect to their ability to reproduce observed streamflow and SC data, their capability of reproducing correctly internal processes and the effect of simplifications in model structure on simulated present and future streamflows. Both approaches are compared, addressing the following questions: (i) How do the models reproduce the temporal and spatial variability of key hydrological processes relevant for simulation of runoff, (ii) what level of effort is required for model calibration and (iii) what are the possibilities of validating key processes with field data and what is the added value of local information for model performance when these data are used to constrain model parameters?

The overarching objective of this study is to assess which modelling approach is appropriate for the estimation of water resource availability for the present and future climate in high Andean watersheds. We attempt to provide evidence about the model complexity and spatial discretization that are necessary for accurate hydrological simulations at different temporal scales, with application to long-term water resource evaluation, including the effect of climate change.

STUDY SITE AND DATA

This study focuses on the Juncal River Basin [33°S, 70°W; 2200–5900 m above sea level (a.s.l.)], a sub-basin of the Upper Aconcagua Basin (Figure 1), located in the Central Andes of Chile (32–36°S). The area of the Juncal River Basin is 241 km². The basin is mainly characterized by a large U-shaped valley filled with Holocene sediments comprising glacio-fluvial deposits and debris flow deposits

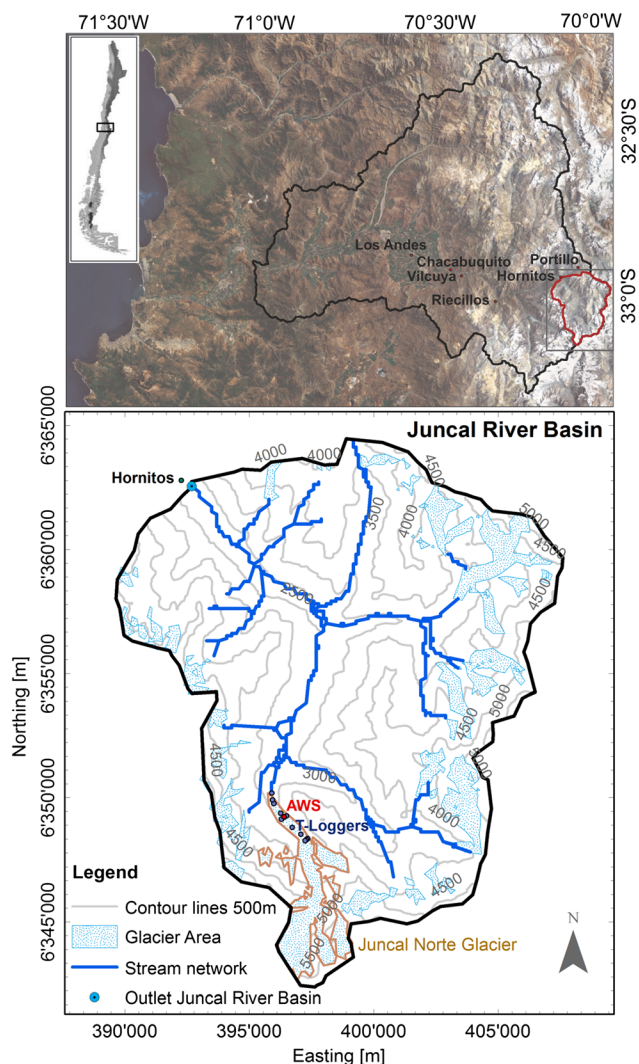


Figure 1. Top: Map of the Upper Aconcagua River Basin. The stations of the standard meteorological network (Los Andes, Chacabuco, Vilcuya, Riecillos and Portillo) are indicated with red dots and names as well as the weather station providing data for snow albedo (Hornitos). The black line indicates the outline of the Upper Aconcagua River Basin closed at La Calera and the red line the margins of the Juncal River Basin. Bottom: Map of the Juncal River Basin with position of the automatic weather station installed on Juncal Norte Glacier during summer 2005/2006 (AWS) and locations of the 2-m air temperature sensors (T-Loggers) installed during summer 2008/2009. Contour lines show 500-m elevation bands

from the surrounding slopes. Neighbouring peaks reach elevations up to 6000 m a.s.l., and several lateral smaller valleys with north–south orientation converge into the main river channel. Sclerophyll vegetation covers soils at elevations up to 2750 m a.s.l. (8% of total area). Bare soils and rocks of volcanic origin are present at higher elevations.

The Aconcagua River Basin is one of the major glacierized basins in Central Chile, with 121 km² of ice in 2003 (Bown *et al.*, 2008). A recent glacier inventory of the region (Bown *et al.*, 2008), providing glacier outlines to this study, reported that the Juncal River Basin contributes by 39.1% to the total glacierized area of the

Aconcagua River Basin. Fourteen per cent of the area of the Juncal River Basin is glacierized. Although the majority of hydrological studies from the region neglect the presence of glaciers because at the regional scale, their relative importance decreases and because the actual contribution of glacier runoff to river flow from headwater catchments is not known, a modelling study by Ragetti and Pellicciotti (2012) showed that the Juncal River Basin has a nivo-glacial hydrological regime. Simulated glacier contribution to total streamflow was 14.4% during the hydrological year 2005/2006 (Ragetti and Pellicciotti, 2012), which we assume as an indication of the order of magnitude of glacier contribution to runoff. A recent isotope end-member mixing analysis by Ohlanders *et al.* (2013) showed that this fraction can be substantially higher in unusually dry years.

Climate

Between the outer tropics and midlatitudes, the climate of the Andean region is dominated by the Pacific anticyclone, which oscillates, allowing for the passage of frontal systems during winter, while blocking these in summer (Vuille *et al.*, 2000; Rutllant and Fuenzalida, 2007; Falvey and Garreaud, 2009). The western side of the Cordillera therefore receives winter precipitation from the moist Pacific air masses, whereas summers are dry (Pellicciotti *et al.*, 2008). The dry season is characterized by low relative humidity (<40%) and stable cloud-free meteorological conditions (Pellicciotti *et al.*, 2008). As a consequence, shortwave radiation is particularly strong (Pellicciotti *et al.*, 2008). The occurrence of El Niño Southern Oscillation (ENSO) phenomena plays an important role in controlling snowfall at this latitude and implies a strong interannual variability: La Niña years are rather dry, whereas El Niño years are associated with higher-than-normal precipitation (Escobar and Aceituno, 1998). Masiokas *et al.* (2006) showed significant correlations between snowpack, annual streamflow volumes and ENSO episodes, and Cortés *et al.*, (2011) showed an important correlation of streamflow timing to the ENSO-related signal for the Central Andes, with warm, wet ENSO episodes related to delayed streamflow volumes in the Central Andes due to a probable thicker snowpack.

Hydrometeorological historical records and DEM

Historical records from a network of hydrometeorological stations run by the Chilean water directorate [Dirección General de Aguas (DGA)] are used to provide model input and calibration data (Table I). We use temperature data from Portillo (3000 m a.s.l.) and precipitation measurements from Riecillos (1290 m a.s.l.) from the period April 2004 until March 2010 to drive the models. Mean annual precipitation in Riecillos was 614 mm with a coefficient of variation of

Table I. Characteristics and location of the hydrometeorological stations used in this study

Station	Elevation (m a.s.l.)	Latitude	Longitude	Location	Period of functioning
Portillo	3000	32°50'28.30"S	70°05'39.18"W	Aconcagua Valley	01 Apr 2004–31 Mar 2010
Riecillos	1290	32°56'01.82"S	70°21'50.15"W	Aconcagua Valley	01 Apr 2004–31 Mar 2010
Vilcuya	1100	32°51'52.37"S	70°28'31.29"W	Aconcagua Valley	01 Apr 2004–31 Mar 2010
Chacabuquito	1030	32°50'49.62"S	70°30'39.56"W	Aconcagua Valley	01 Apr 2004–31 Mar 2010
Los Andes	830	32°48'27.20"S	70°38'19.79"W	Aconcagua Valley	01 Apr 2004–31 Mar 2010
AWS1	3127	32°59'26.58"S	70°06'31.27"W	Juncal Norte Glacier tongue	11 Dec 2005–12 Feb 2006
Hornitos	2218	32°52'4.16"S	70°9'5.48"W	Aconcagua Valley	01 Feb 2010–30 Oct 2011
Stream gauge	2233	32°52'11.17"S	70°08'57.96"W	Juncal River Valley	01 Apr 2004–31 Mar 2010 ^a

^a With some missing data in 2007 and 2008.

44%. Mean annual temperature in Portillo was 4.8 °C, with a minimum mean monthly temperature in August of −0.8 °C and a maximum mean monthly temperature in January of 11 °C. Data from Vilcuya (1100 m a.s.l.), Chacabuquito (1030 m a.s.l.) and Los Andes (830 m a.s.l.) provide additional information required for precipitation and temperature regionalization (Section on Extrapolation of Meteorological Input Data). All stations providing meteorological data are located in the main valley of the Upper Aconcagua River Basin (Figure 1). The meteorological network near the watershed is therefore relatively dense, but spatial coverage is limited to lower elevations. The sparseness of the hydrometeorological network at high elevations is common to the entire region and prevents a more detailed analysis of the spatial and temporal patterns of climate in the region (Liboutry, 1998; Masiokas *et al.*, 2006).

The period covered by the available hydrometeorological data includes both cold and warm ENSO years. The multivariate ENSO Index (MEI, Wolter and Timlin, 1993, 1998, available at <http://www.esrl.noaa.gov/psd/enso/mei/table.html>) can be used to classify hydrological years in cold and warm ENSO episodes: Periods with positive MEI values (warm years) are the hydrological years 2004/2005, 2006/2007 and 2009/2010, whereas 2007/2008 was a year with particularly negative MEI values (cold year).

Daily values of streamflow data are provided by a gauge station at the outlet of the Juncal River Basin, also maintained by the DGA. Water levels are recorded by radar and then converted into runoff using rating curves periodically derived by the DGA. For the period of record of this study, five curves were used (derived at intervals of approximately 1 year).

The Shuttle Radar Topography Mission 90-m resolution digital elevation model (DEM) is used to discretize elevation bands in WEAP and to attribute grid cell elevations in TOPKAPI-ETH. Slope and aspect as well as the channel network in TOPKAPI-ETH are calculated on the basis of the DEM, which is used also for calculation of the clear-sky solar radiation and interaction with topography (Section on TOPKAPI-ETH).

MODIS and Landsat snow cover

We use observations about the extent of SC in the region, acquired by the Moderate-resolution Imaging Spectroradiometer (MODIS) optical instrument mounted on the Terra and Aqua satellites of the National Aeronautics and Space Administration Earth Observation System. Like a variety of geophysical products derived from MODIS observations, the global SC product is freely available through the Distributed Active Archive Center of the National Snow and Ice Data Center (www.nsidc.org). MODIS SC products are provided both on a daily basis and as 8-day composites at 500-m resolution over the Earth's land surfaces (Hall *et al.*, 2006). The 8-day MOD10A2 product contains data fields for maximum SC extent over an 8-day period. The statistical difference between daily and 8-day MODIS images over a period of 10 years is small (J.-P. Dedieu, personal communication), as the latter is less affected by clouds; the lower processing requirement associated to the 8-day composites makes them suitable for this study.

In the MODIS data, grid cells are classified as snow (1) or no snow (0). See Hall and Riggs (2007) for an accuracy assessment of the MODIS snow products and a description of the MODIS snow mapping algorithm. We calculate the percentage of snow-covered area (SCA) over the study catchment observed by MODIS and compare it with the maximum 8-day snow extent given by the models (snow water equivalents are converted to SCA if they are >0). Average cloud cover on 8-day MODIS images during our study period was 4.2%, with a seasonal maximum from April to June of on average 12.8%.

The MODIS daily or 8-day snow maps have been successfully applied in the Central Andes for validation of modelled snow accumulation and snowmelt dynamics by Gascoin *et al.* (2013) and by Stehr *et al.* (2009). MOD10A2 data are used in this work for model calibration (WEAP) and validation. However, for analysis of the model performance at high elevations with rough topography and to evaluate the importance of snow redistribution, we also use Landsat 5

fractional SC. SC in mountainous terrain retrieved from the finer but less frequent Landsat images is presumably more accurate than SC retrieved by the standard MODIS SC algorithm (Rittger *et al.*, 2013). Landsat 5 has a spatial resolution of 30 m and 16-day overpass. Not all overpasses are available for the study region. We used four images not affected by clouds (9 Apr 2006, 11 Mar 2007, 3 Mar 2008 and 16 Mar 2009) and calculated the fractional SCA using a slightly modified version of the Rosenthal and Dozier (1996) fractional SCA regression trees. The error of these SCA calculations is in the order of 7% (Rosenthal and Dozier, 1996).

Local meteorological data

Data collected by an automatic weather station (AWS) installed by an ETH expedition (Pellicciotti *et al.*, 2008) on the Juncal Norte Glacier during summer 2005/2006 are used for determination of TOPKAPI-ETH model parameters, as described by Ragetti and Pellicciotti (2012). Hornitos, a meteorological station located 500 m downstream of the study catchment's outlet (Figure 1), provides winter incoming and reflected shortwave radiation (W m^{-2}) not available at the AWS but required for the estimation of fresh snow albedo in TOPKAPI-ETH. The station is run by the Civil Engineering Department, University of Chile, since 2010.

Eleven temperature sensors (T-Loggers) installed at 2 m above the surface on the glacier tongue of the Juncal Norte Glacier and one temperature sensor installed near it (Figure 1) between 8 December 2008 and 14 February 2009 provide additional air temperature data to assess the value of detailed local information for model performance. Mean air temperatures measured by the T-Loggers and in Portillo are shown in Figure 2. The dataset is discussed in detail by

(Petersen and Pellicciotti, 2011), who found that temporally variable lapse rates improved the representation of temperature variability over the Juncal Norte Glacier. The reader is referred to that publication for a detailed description of the Juncal Norte Glacier air temperature regime.

METHODS

Extrapolation of meteorological input data

Stations belonging to the standard meteorological network are located outside the limits of the study catchment (Figure 1). Meteorological input data are thus extrapolated to our study site. We calculate precipitation gradients by comparing annual precipitation amounts at the Riecillos (1290 m a.s.l.) and Portillo (3000 m a.s.l.) stations. In the case of temperature, lapse rates are calculated on the basis of daily mean temperatures at Vilcuya (1100 m a.s.l.) and Portillo. Average lapse rates calculated for the area are shown in Figure 3.

Lapse rates are assumed to be linear in the case of temperature and logarithmic in the case of precipitation (Figure 3). Observations suggest that the Andes exerts a strong influence on the precipitation regime of this region, resulting in increased precipitation with elevation (Falvey and Garreaud, 2009). However, the lack of high-elevation climate stations prevents a detailed analysis of the spatial and temporal patterns of precipitation over the Central Andes (Falvey and Garreaud, 2009). The logarithmic lapse rate was chosen because linear precipitation gradients usually overestimate the total volume of precipitation when compared with observed total runoff (Vicuña *et al.*, 2010). Logarithmic lapse rates also reflect more accurately the orographic forcing effect typical of

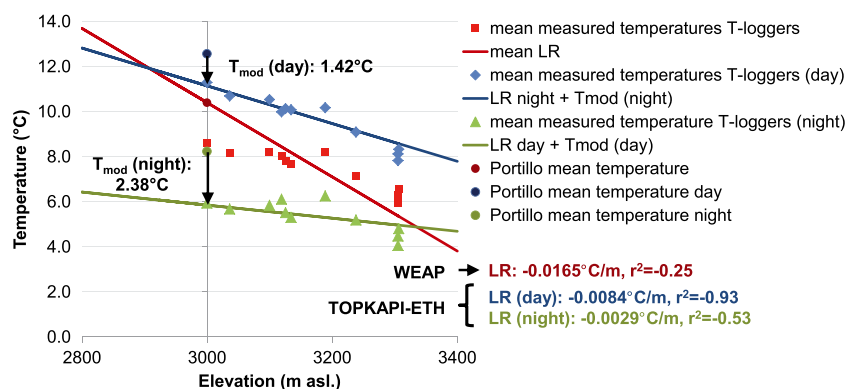


Figure 2. Observed air temperature distribution on Juncal Norte Glacier, at Portillo and air temperature lapse rates used for extrapolation. The observed temperatures represent the mean values at the locations of the T-Loggers and in Portillo, for the period 8 Dec 2008–12 Feb 2009. Temperature lapse rates (LR) and temperature decrease over glacierized areas (T_{mod} , black arrows) are calculated by fitting extrapolated temperatures from Portillo (3000 m above sea level) to observed temperatures. r^2 is the coefficient of determination between observed and extrapolated temperatures. The green and blue lines represent the extrapolated temperatures by TOPKAPI-ETH, and the red line the extrapolated temperatures by WEAP (LR configuration 3, Table IV)

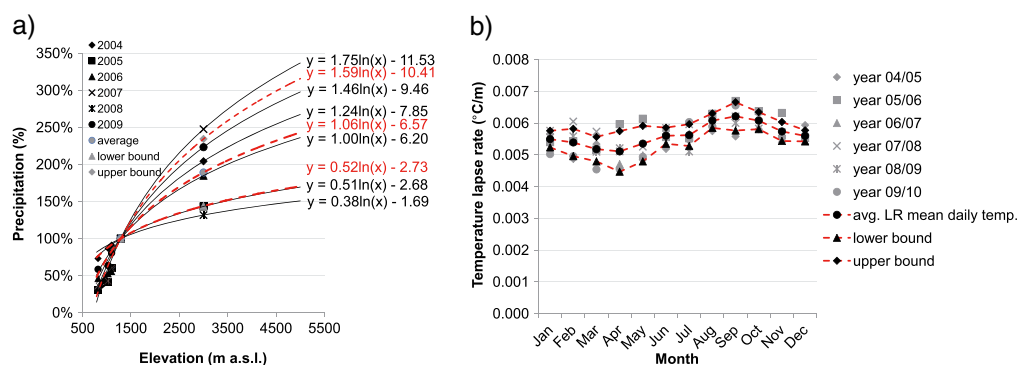


Figure 3. (a) Precipitation gradients calculated on the basis of data from Riecillos [1290 m above sea level (a.s.l.)] and Portillo (3000 m a.s.l.). Precipitation data from Vilcuya (1100 m a.s.l.), Chacabuquito (1030 m a.s.l.) and Los Andes (830 m a.s.l.) are also shown, as well as the equations used for extrapolation of daily precipitation; (b) temperature lapse rates (LR) calculated on the basis of Vilcuya and Portillo. For temperature, we use linear lapse rates to extrapolate measured data. Upper/lower bounds correspond to mean lapse rates plus/minus one standard deviation in the observed lapse rates

high elevations, by which wet air masses are orographically lifted and discharge their humidity content in correspondence to the mountain rises. Extrapolated precipitation is compared with observed precipitation at the five stations in the region including the lower-laying stations in the Aconcagua River valley: Vilcuya and Chacabuquito at 1030 m a.s.l. and Los Andes at 830 m a.s.l. (Figure 3a). The coefficient of determination is higher if a logarithmic lapse rate is used ($r^2=0.71$) than if a linear lapse rate is used ($r^2=0.62$).

Model description

TOPKAPI-ETH. TOPKAPI-ETH is a fully distributed rainfall–runoff model that has been developed for applications to mountainous basins. The version used in this study is almost identical to the version used by Ragetti and Pellicciotti (2012) and Finger *et al.* (2011, 2012). Water routing in TOPKAPI-ETH as well as in its precursors is based on the kinematic wave concept, whereby soil drainage, overland flow and channel flow are represented by nonlinear reservoir differential equations (Liu and Todini, 2002, 2005), which are resolved for each catchment grid cell.

Snowmelt and glacier melt are computed using an enhanced temperature index (ETI) approach (Pellicciotti *et al.*, 2005, 2008), where melt in each grid cell is a function of a temperature-dependent term representative of the temperature-dependent energy fluxes (and equal to a constant empirical temperature factor, TF , $\text{mm h}^{-1} \text{ } ^\circ\text{C}^{-1}$, multiplied by ambient air temperature) and a shortwave radiation dependent term that uses the fully distributed shortwave radiation balance. Changes in glacier areas are not considered for this study, which is a reasonable assumption for simulations over only few years for this study region (Bown *et al.*, 2008).

Potential clear-sky global irradiance is simulated with a nonparametric model based on the work of Iqbal (1983), accounting for the position of the sun relative to the

considered grid cell at each time step. The vectorial algebra approach proposed by Corripio (2003) is used for the interaction between the solar beam and terrain geometry. Transmission through the atmosphere is taken into account using transmissivity coefficients described by Pellicciotti *et al.* (2011). Incoming shortwave radiation is then the product of the clear-sky irradiance and a cloud transmittance factor derived from the range of diurnal variations of air temperature (Pellicciotti *et al.*, 2011).

The equivalence between net shortwave radiation (W m^{-2}) and units of melt ($\text{mm water equivalents h}^{-1}$) is controlled by a shortwave radiation factor (SRF). The magnitude of the empirical factors in the enhanced temperature index model can be related to the surface energy balance typical of the climatic setting (Pellicciotti *et al.*, 2008; Carenzo *et al.*, 2009; Pellicciotti *et al.*, 2012; Ragetti and Pellicciotti, 2012). Snow albedo is estimated as a function of an empirical parameter α_2 that controls the logarithmic decrease of albedo and the maximum albedo after snowfall (α_1) (Brock *et al.*, 2000). Ice albedo (α_{ice}) is assumed constant (Pellicciotti *et al.*, 2008).

The Makkink (1957) approach, based on air temperature and incoming shortwave radiation, is used to compute potential evapotranspiration. Actual evapotranspiration depends on available soil moisture content within a superficial soil layer calculated internally by the model. A second soil layer accounts for runoff originating from percolation to deeper soil and into fractured bedrock (Finger *et al.*, 2011). Spatially homogeneous characteristics are attributed to this second soil layer because no information on the spatial extent of groundwater aquifers is available.

In this study, TOPKAPI-ETH is applied at an hourly time step and a grid resolution of 100 m. Table II provides an overview of the most important model characteristics and represented processes. For more detailed information about TOPKAPI-ETH and complete lists of model

Table II. Main characteristics of the two models (WEAP and TOPKAPI-ETH) evaluated in this study

	WEAP	TOPKAPI-ETH
1. Spatial representation	Semi-distributed (100-m elevation bands)	Fully distributed (100 × 100-m grid cells)
2. Temporal resolution	Daily	Hourly
3. Precipitation distribution	Station data, extrapolation with logarithmic gradients (Figure 2a)	Station data, extrapolation with logarithmic gradients (Figure 2a)
4. Air temperature distribution	Station data, extrapolation with linear lapse rates (Figure 2b)	Station data, extrapolation with linear lapse rates (Figure 2b)
5. Solar radiation	Single lumped parameter, depending on the day of the year and latitude, no effect of topography nor transmission through the atmosphere	Specific temperature distribution on glaciers (Figure 3) Fully distributed shortwave radiation taking into account • Hourly clear sky global irradiance (Iqbal, 1983) and transmission through the atmosphere, partition between direct and diffuse radiation • Terrain geometry (Corripio, 2003) and interaction with topography (shading and multiple reflection) • Daily cloud transmittance factors (Pellicciotti <i>et al.</i>, 2011)
6. Snow albedo	Linear decrease with time	Logarithmic decrease as a function of maximum daily positive air temperatures (Brock <i>et al.</i> , 2000)
7. Snowmelt	Melt coefficient (Yates <i>et al.</i> , 2005), maximum daily snowmelt limited by a lumped radiation parameter	Enhanced temperature index approach (Pellicciotti <i>et al.</i> , 2005, 2008)
8. Gravitational snow redistribution	—	Snow redistribution depending on a slope threshold (β)
9. Glacier melt	—	Enhanced temperature index approach (Pellicciotti <i>et al.</i> , 2005, 2008)
10. Evapotranspiration	Potential evapotranspiration: simplified Penman–Monteith (Yates <i>et al.</i> , 2005), depending on • Air temperature	Potential evapotranspiration (Makkink, 1957), depending on • Crop factors • Air temperature • Incoming shortwave radiation
11. Discharge routing	Actual evapotranspiration depending on • Relative catchment storage state (linear reservoir) Two-bucket system for each 100-m elevation band	Actual evapotranspiration depending on • Available soil moisture Kinematic wave concept (Liu and Todini, 2002; Liu and Todini, 2005) for modelling channel, soil and surface flow
12. Soil representation	Two-bucket system for each 100-m elevation band	Two soil layers and physical properties

parameters, we refer to the publications of Finger *et al.* (2011) and Ragetti and Pellicciotti (2012). The latter applied the model to the same basin as in this study.

WEAP. Water Evaluation and Planning is a water resources system analysis tool developed by the Stockholm Environment Institute (Yates *et al.*, 2005) that integrates a water management model driven by water demands, supplies and environmental requirements with a hydrological model based on conceptual rainfall–runoff relationships. The conceptual hydrological representation of a watershed is based on a two-bucket system: The first bucket represents a soil layer that reproduces the short-lag watershed response to precipitation, and the second layer is designed to mimic the baseflow or longer-lag flows due to the presence of deep soils or aquifers within the catchment (Winter, 2001; Burness, 2004). The watershed is divided into fractional areas, where the equations governing water movement are solved. For this study, 100-m elevation bands are used to assign fractional areas. This paper uses only the hydrological tool of the WEAP model, because human water use within the study area is negligible and the evaluation of impacts of hydrological changes on the water resource system is not a focus of this study.

Each soil layer is assigned quasi-physical properties such as total storage capacity, conductivity and preferential flow direction. These properties can be defined for each fractional area and are tuning parameters of WEAP (Yates *et al.*, 2005). Daily time steps are used in this study to calculate soil moisture in the function of effective precipitation, surface runoff, subsurface runoff and baseflow. The reader is referred to Yates (1996) for a description of functional forms of flow components.

Evapotranspiration is a function of potential evapotranspiration and the relative catchment storage state. Potential evapotranspiration is calculated from a simplified version of the Penman–Monteith method as provided by the Food and Agriculture Organization manual on evapotranspiration, in which energy fluxes are calculated as a function of air temperature only (Yates *et al.*, 2005).

Liquid precipitation and meltwater result in what is defined effective precipitation. Both components are calculated in the function of a melt coefficient, which accounts for melting and freezing temperature thresholds (two tuning parameters). The melt coefficient is 0 if ambient air temperature is below freezing temperature and 1 if air temperature is above melting temperature. If air temperature is between freezing and melting temperature, the melt coefficient is linearly interpolated (Yates *et al.*, 2005). The maximum amount of the accumulated snowpack that can be turned into meltwater during each time step is limited by an empirical, lumped parameter representing the available melt energy (Young *et al.*, 2009) (units in water equivalents depth per time). This is calculated on the basis

of a solar radiation index (which takes into account only the day of the year and the latitude, but not shading, interaction with topography, transmissivity through the atmosphere and cloud effects) and snow albedo decreasing linearly with time.

The present version of WEAP has been used in numerous water resource management applications in various regions of the world, as well as in our study region for climate change impact studies (Vicuña *et al.*, 2010; Cortés *et al.*, 2012; Poblete *et al.*, 2012). For a complete list of previous applications of WEAP, the reader is referred to the WEAP21 official website (www.weap21.org).

Model calibration

Usually, it is supposed that physically based hydrological models require less calibration, because model parameters can be estimated directly from measurements or from the literature (Koren *et al.*, 2004; Kuzmin *et al.*, 2008; Zhang *et al.*, 2012). The opposite applies with conceptual models, for which parameters correspond less directly to physical properties. The choice of calibration strategies therefore depends strongly on model structure. The calibration strategies designed for this work reflect this issue: For WEAP, we choose a manual direct calibration method, whereas we use local data (Section on Local Meteorological Data) for an estimation of the physically measurable TOPKAPI-ETH model parameters.

We divide the 6-year record in three subperiods: (i) 1 year of data is required for the generation of initial conditions of soil moisture and SC (hydrological year 2004/2005), (ii) 2 years for calibration (April 2005–March 2007) and (iii) 3 years for validation (April 2007–March 2010).

For WEAP calibration, model parameters are divided into two groups, depending on their relation to (i) processes related to generation of snowmelt (processes 5–7 in Table II) and (ii) processes more directly related to streamflow generation such as meltwater routing or soil moisture (processes 10–12 in Table II). The two groups of parameters are then calibrated manually by first fitting simulated to observed MODIS SC and, secondly, by fitting simulated to measured streamflow at the outlet of the Juncal River Basin. The stepwise calibration guarantees a minimum of internal consistency and reduces the risk of error compensation (Huss *et al.*, 2009; Pellicciotti *et al.*, 2012; Ragetti and Pellicciotti, 2012). Parameters governing the extrapolation of meteorological input variables are excluded from the calibration scheme, and data from the standard meteorological network are used to estimate lapse rates of temperature and precipitation (Section on Extrapolation of Meteorological Input Data, Figure 3).

Two main problems exist for setting up TOPKAPI-ETH without calibration. First, not all model parameter values can be directly estimated by measurements in the field. Complex

processes, such as evapotranspiration or the redistribution of snow by wind and gravity, cannot be fully described by a model at a 100-m grid resolution by relying on only temperature and precipitation as input. Such processes need to be parametrized, and the corresponding model parameters need to be calibrated using specific data. Second, parameter values in distributed models often vary in space, so that for physically based models, the parameters' spatial variability has to be taken into account by regionalization, if parameters are estimated from local observations. Particularly, in the case of headwater catchments that reach high altitudes, spatial interpolation or extrapolation of model parameters would not be supported because of extreme elevation gradients or low correlation lengths.

In order to alleviate this problem, we use observed field data for a direct estimation of model parameters whenever possible. Table III shows the model parameters estimated from different point-scale data. The melt parameters TF , SRF and T_T are taken from Ragettli and Pellicciotti (2012), where they were determined against ablation rates simulated by an energy balance model (Pellicciotti *et al.*, 2008) forced with data from AWSs installed on Juncal Norte Glacier during summer 2005/2006. Ragettli and Pellicciotti (2012) also used data from a second ETH expedition in summer 2008/2009. However, data and parameter values from this period are not considered because they do not overlap with the calibration period used in this study. Extrapolated air temperatures (Section on Extrapolation of Meteorological Input Data) are compared with temperature measured at the AWS on Juncal Norte Glacier during summer 2005/2006: The observed

mean difference is used as a constant temperature-decreasing factor (T_{mod}) over glacierized areas. This accounts for the fact that temperatures in the glacier boundary layer are lower than those outside (Greuell and Böhm, 1998; Shea and Moore, 2010; Petersen and Pellicciotti, 2011). Because of the absence of precipitation during summer, the AWS could not provide an estimate of fresh snow albedo (α_1), but data from the Hornitos valley station (Figure 1) were used. The parameter controlling the logarithmic decay of fresh snow albedo as a function of cumulated daily maximum positive air temperature (α_2) is calibrated using point observations of albedo from the AWS, as was carried out by Ragettli and Pellicciotti (2012). For the threshold temperature to distinguish between solid and liquid precipitations, we assume a constant value of 1 °C (Stehr *et al.*, 2009). The threshold slope for snow gravitational redistribution (β_{MAX}) is taken from Ragettli and Pellicciotti (2012), where it was calibrated with measurements of snow depth. Because of the lack of more detailed information, we assume model parameters related to snow accumulation and snowmelt and ice melt to be constant in time and uniform in space. This assumption is supported by the results of Ragettli and Pellicciotti (2012).

Parameters controlling soil-related and evapotranspiration-related processes cannot be estimated directly using point-based observations, because these processes vary considerably in space and because data were not available. Considering plausible ranges of parameter values (see Finger *et al.*, 2011, for a complete list of TOPKAPI-ETH parameters and plausible soil parameter values), soil-related and evapotranspiration-related parameters are therefore

Table III. Point-scale estimation of TOPKAPI-ETH model parameters

Parameter	Units	Value	Source	Description
Temperature distribution				
LR	°C m ⁻¹	Figure 3	VIL and POR	Linear temperature lapse rate representing the decrease of temperature with elevation
T_{mod}	°C	2.86	AWS	Temperature decrease over glacierized surface
Precipitation distribution				
P_{grad}	—	Figure 3	RIE and POR	Logarithmic precipitation gradient representing the change of precipitation with elevation
Snowmelt and ice melt				
TF	mm day ⁻¹ °C ⁻¹	0	AWS	Temperature factor
SRF	m ² mm ⁻¹ W ⁻² h ⁻¹	0.0105	AWS	Shortwave radiation factor
T_T	°C	6	AWS	Threshold air temperature for melt onset
Albedo				
α_1	—	0.83	HOR	Albedo of fresh snow
α_2	—	0.112	AWS	Decay of snow albedo
α_{ice}	—	0.165	AWS	Albedo of ice (glacier surface)

TOPKAPI-ETH parameters shown in this table are not calibrated but estimated from point-scale data. See Section on TOPKAPI-ETH for more information on melt and albedo parameters.

HOR, Hornitos; VIL, Vilcuya; POR, Portillo; RIE, Riecillos; AWS, automatic weather station installed on Juncal Norte Glacier during summer 2005/2006.

calibrated manually against runoff measured at the outlet of Juncal River Basin. Soil maps were not available for the area, but we constrain the range of possible parameter values by assuming that soil depth decreases with elevation and slope and that saturated conductivity increases with elevation and slope in order to account for the presence of fractured rock systems instead of soils at higher elevations.

Model validation

For model validation, we assess how the models perform when using an average lapse rate over several years, or a more detailed dataset of annual lapse rates (validation steps 1 and 2 in Table IV). In this way, we benefit from a higher temporal resolution that captures the variability of meteorological conditions. This is of particular interest because ENSO phenomena play an important role in modulating annual precipitation volumes in this region (Escobar and Aceituno, 1998; Rivera *et al.*, 2000; Masiokas *et al.*, 2006).

In a third step (validation step 3, Table IV), we validate WEAP and TOPKAPI-ETH using local information about temperature lapse rates and the temperature decrease over glacierized areas calculated on the basis of detailed field data from Juncal Norte Glacier (Figure 1) from the summer 2008/2009 (Section on Local Meteorological Data). Because no additional local precipitation records exist, precipitation gradients are the same as in the validation step with annually varying gradients.

The three validation steps provide an estimate of the value of detailed, local information in data-scarce regions for different types of hydrological models. They also answer the question of whether additional meteorological information acquired after calibrating each model can be effectively used to improve model performance.

In order to make maximal use of the available dataset, each model uses the available information differently. We therefore account for the fact that the value of information depends also on model structure. TOPKAPI-ETH uses different temperature lapse rates and on-glacier temperature-decreasing factors (T_{mod}) during daytime and night-time

(Petersen and Pellicciotti, 2011). As WEAP performs calculations at daily time steps, the lapse rate has to be calculated on the basis of mean daily temperatures (Figure 2).

Analysis of process representation

In order to compare process representation during different periods between the two models, model outputs are screened through a principal component analysis (PCA, Legendre and Legendre, 1998). In a PCA, a cloud of points (observations or model results), situated in the initial space of descriptors (variables), is transposed to a reduced space where the similarities among points are preserved and represented on a reduced number of axes. The basis for the projection on the reduced number of axes is the correlation matrix whose elements are the cross-correlations between different standardized model variables. If all variables are equally well represented in a two-dimensional reduced space, their corresponding descriptor axes can be represented by arrows in a circle, where the cosine of an angle between two arrows represents the correlation between variables.

If we use relevant water balance components as variables, a PCA allows a direct comparison of the temporal and spatial variability of dominant internal hydrological processes among models and provides insights about temporal and spatial differences in process representation, which lead to differences in model outputs (discussed in Section on Process Representation). In comparison with conventional plotting methods, it offers the advantage that all process components can be visualized together.

Sensitivity to temperature increase

Climate models project important future changes in meteorological conditions for the western coast of South America, especially for temperature (Bradley *et al.*, 2004). We assess how the models react to a warmer climate by adding a constant increase, ΔT (different for each month of the year), to the historical input air temperatures. Although our approach is to test a simplified temperature increase scenario and not drive the models with direct global climate model (GCM) outputs, we attempt to choose a realistic one

Table IV. Lapse rate configurations for calibration and validation

Lapse rate configurations	
Calibration	Constant lapse rates based on meteorological information from standard meteorological network, measuring period 01/04/2005–31/03/2007 (lapse rate configuration 1)
Validation step 1	Constant lapse rates based on meteorological information from standard meteorological network, measuring period 01/04/2007–31/03/2010 (lapse rate configuration 1)
Validation step 2	Annual lapse rates based on meteorological information from standard meteorological network (lapse rate configuration 2)
Validation step 3	Temperature lapse rates calculated on the basis of local temperature measurements, precipitation gradients as in validation step 2 (lapse rate configuration 3)

on the basis of findings from recent works. The temperature scenario applied for this study is the ensemble mean of projections by ten GCMs (A2 scenario) for the period 2045–2065, downscaled to the El Yeso meteorological station (2475 m a.s.l., 70 km south of Juncal) by Cortés *et al.* (2012). We refer to this study for more details about the downscaling procedure and the GCMs. The temperature change projections are highly variable among different seasons: Whereas spring and summer (August–February) temperatures are increasing on average by 2.25 °C, autumn and winter (March–July) temperatures are on average increasing by 0.75 °C only (see Table V for monthly ΔT values).

Recent studies have shown that temperatures will increase more in the high mountains than at lower elevations (Bradley *et al.*, 2006; Rangwala *et al.*, 2009; Urrutia and Vuille, 2009). This effect is taken into account by using shallower lapse rates than for the present for the simulations with the future temperature scenario, so that temperature warming is higher for high elevations. Bradley *et al.* (2006) suggested a warming of about 0.4 °C per 1000-m elevation difference for the Andes at 33°S, the same as that suggested by Urrutia and Vuille (2009) for the western slope of the Andes. This corresponds to 0.0004 °C m⁻¹ shallower temperature lapse rates than for the present (Table V).

To initialize the models for the future conditions, we run the models repeatedly with present conditions, until the snowpack at places where the two models accumulate snow under present climatic conditions is sufficiently thick so that, subsequently, the simulated response to a temperature increase is independent of the initial snow conditions.

In order to compare the sensitivity of the models with a temperature increase, we calculate 3-month average

values of model outputs for different variables and plot the results using a non-metric multi-dimensional scaling (NMDS) representation (Legendre and Legendre, 1998). This scaling method preserves the similarity among model results for each 3-month period in a reduced number of dimensions on the basis of the Euclidean distance matrix. The clustered representation of 3-month periods that react similarly to temperature increase creates a map of similarities, which allows an efficient identification of differences among models, among variables and among 3-month periods.

RESULTS

Model calibration

Results for the calibration period show that both models are able to reproduce adequately 8-day maximum SC and runoff (Table VII and Figures 4 and 5): Coefficients of determination (r^2) are about 0.8 in the case of modelled SC, and Nash and Sutcliffe efficiency (NSE) values are about 0.9 for simulated runoff.

Maximum winter SC simulated by TOPKAPI-ETH is lower than MODIS SC throughout the winter (Table VII and Figure 5a). This can be explained by the higher grid resolution of TOPKAPI-ETH with respect to MODIS images (500 m) and the fact that steep mountain slopes are never completely snow covered because of the effect of wind and gravity (Section on Snow Accumulation). Minimum summer SC simulated by TOPKAPI-ETH agrees very well with observed MODIS SC (Table VII). WEAP, on the other hand, tends to underestimate summer SC (January–March values in Figure 5). Both models agree well with observed SC from October to December, which is the main melting period in the southern hemisphere. Simulated SC from short-term events such as summer storms (January–March) and the onset of winter (April–June) shows generally less agreement with MODIS (Figure 5). However, the temporal resolution of 8-day maximum SC might not be sufficient to detect precisely such events.

Given that MODIS SC was not used for parameter calibration in TOPKAPI-ETH, our case study demonstrates that local information can be efficiently used for calibration of the cryospheric components of a distributed, physically oriented model.

Both models deliver high efficiencies with regard to simulated daily runoff in the calibration period (Table VII and Figure 4). The hourly TOPKAPI-ETH-simulated streamflow was averaged to daily values to compare it with available observations and WEAP daily runoff outputs. In both hydrological years used for calibration, observed runoff reaches its maxima by the end of December/beginning of January and then decreases until

Table V. Scenario applied for the test of the models' sensitivity to a temperature increase

	ΔT (°C)	ΔLR (°C/m)	LR (°C/m)
Jan	2.32	0.0004	– 0.0051
Feb	2.14	0.0004	– 0.0050
Mar	1.2	0.0004	– 0.0048
Apr	0.37	0.0004	– 0.0048
May	– 0.14	0.0004	– 0.0050
Jun	0.87	0.0004	– 0.0051
Jul	1.46	0.0004	– 0.0052
Aug	2.03	0.0004	– 0.0057
Sep	2.25	0.0004	– 0.0058
Oct	1.94	0.0004	– 0.0057
Nov	2.61	0.0004	– 0.0053
Dec	2.47	0.0004	– 0.0052

ΔT are the projections of temperature changes used in this study (based on Cortés *et al.*, 2012). These values are added to the historical data from Portillo. ΔLR are the considered projections of enhanced warming at higher elevations (based on projections reported by Bradley *et al.* (2006) and Urrutia and Vuille (2009)). LR are the monthly lapse rates that are applied to extrapolate future air temperature, corresponding to the average temperature lapse rates between Vilcuya and Portillo (Figure 3) plus ΔLR .

baseflow conditions are reached at the beginning of April (Figure 4). The peak in simulated discharge in mid-February 2006 is due to a summer storm that was recorded at the meteorological station in Riecillos. This peak is not observed in the measurements at the Juncal River Basin outlet, and MODIS data do not exhibit any increase in SC, suggesting that the storm was likely less strong in the Juncal area than in Riecillos. The main peak in runoff observed in January 2007 is not due to a storm event but to a combination of high temperatures, snowmelt and probably saturated soils. Such particular conditions are slightly better reproduced by the distributed model, whose maximum daily runoff in 2007 is closer to that measured than WEAP's (measured: $36.8 \text{ m}^3 \text{ s}^{-1}$, TOPKAPI-ETH: $23.9 \text{ m}^3 \text{ s}^{-1}$, WEAP: $18.7 \text{ m}^3 \text{ s}^{-1}$; Figure 4).

Model validation

Results for the validation period are shown in Figures 6 and 7. First, constant precipitation gradients and temperature lapse rates over the validation period are applied. Table VII shows that the goodness of fit for simulations of both runoff and SC decreases considerably, especially for WEAP, with respect to the calibration period (NSE decreases to 0.52 in the case of TOPKAPI-ETH and 0.17 in the case of WEAP). WEAP overestimates runoff in the summer 2008/2009 and underestimates it in 2009/2010 (Figure 6). TOPKAPI-ETH also overestimates runoff in the summer 2008/2009 and overestimates the daily runoff variability during 2009/2010

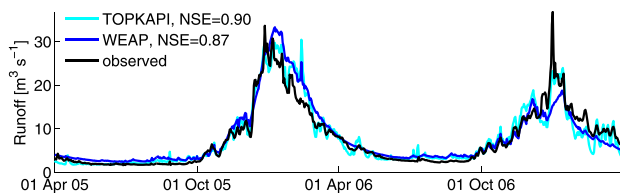


Figure 4. Observed and simulated daily streamflow, Juncal River Basin, calibration period (01/04/2005–31/03/2007). TOPKAPI-ETH hourly outputs are averaged to mean daily values. NSE, Nash and Sutcliffe efficiency;

(Figure 6). No runoff measurements exist for the period November 2007–April 2008.

To assess the effect of possible error sources on model performance, we analyse the influence of different temperature and precipitation lapse rates. We run the models with the mean precipitation and temperature lapse rates (Figure 3) and with the mean lapse rates plus/minus one standard deviation of the observed lapse rates (upper and lower bounds in Figure 3) over the course of 6 years (April 2004–March 2010). Variable precipitation gradients have a strong effect on simulated runoff: WEAP runoff varies by nearly $\pm 30\%$ with respect to the reference model run with mean gradients, whereas TOPKAPI-ETH runoff varies by more than $\pm 20\%$ (Table VI). Mean SC differs by 4–5% from the reference run if upper/lower bounds are applied (Table VI). The effect of the variability in temperature lapse rates on runoff and SC is comparably small: The changes in runoff for both models stay below $\pm 5\%$ (Table VI), with TOPKAPI-ETH showing a larger response to changes in the temperature lapse rate. The effect on simulated SC extent of the given uncertainty in temperature lapse rates is about $\pm 1.5\%$ for both models.

Both models achieve higher goodness of fit for runoff if annual lapse rates are applied (TOPKAPI-ETH: $NSE = 0.69$, WEAP: $NSE = 0.77$; Table VII). The two hydrological years 2007/2008 and 2009/2010 exhibited higher-than-average precipitation gradients, whereas 2008/2009 was characterized by an extremely low precipitation increase with elevation (Figure 3). Accordingly, simulated runoff increases with respect to the validation step 1 for 2007/2008 and 2009/2010 and decreases for 2008/2009. Although better estimates of precipitation gradients lead to an increase of model efficiency with respect to runoff, model agreement with MODIS SC is practically unaffected.

For TOPKAPI-ETH, model performance increases further by using temperature lapse rates calculated on the basis of local air temperature measurements (validation step 3 in Table IV) with NSE values as high

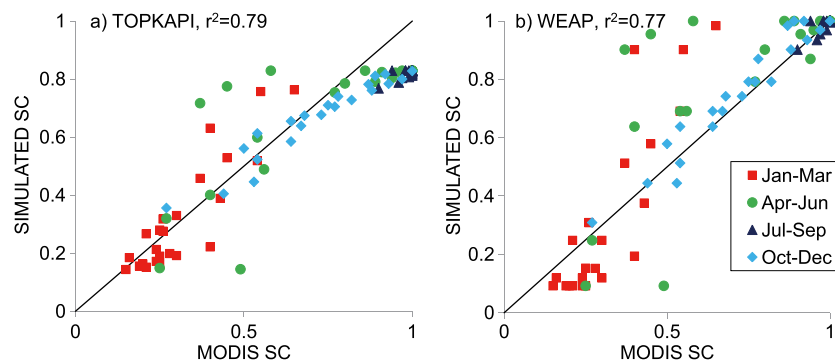


Figure 5. MODIS observed snow cover (SC) and snow cover simulated the models TOPKAPI-ETH (a) and WEAP (b), Juncal River Basin, calibration period (01/04/2005–31/03/2007). r^2 is the coefficient of determination between observed and simulated snow cover

as 0.88 (Table VII and Figure 6). On the other hand, the agreement with MODIS SC decreases slightly with respect to the previous validation step: r^2 decreases from 0.76 to 0.70 and the annual minimum SC is overestimated by 2.8% of the catchment area (Table VII). This is a small value, but it might indicate that the locally observed air temperature lapse rates might be representative of conditions at high elevations and over glacierized parts of the basin, but less of the temperature distribution at low elevations.

For WEAP, results of the third validation step are not shown because the model could not make efficient use of this dataset. The structure of WEAP does not allow taking into account the cooling effect of cold surfaces of snow or ice. For this reason, the coefficient of determination of temperatures extrapolated to the Juncal Norte Glacier on the basis of daily data from Portillo is very low ($r^2 = 0.25$,

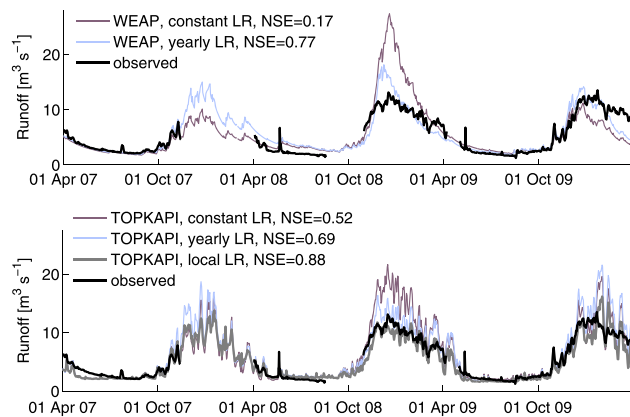


Figure 6. Observed runoff (black line) and runoff simulated by the two models in the validation period with meteorological lapse rates (LR) of increasing temporal resolution: constant precipitation and temperature lapse rates (constant LR), annually varying precipitation and temperature lapse rates (yearly LR), local information about temperature lapse rates from Juncal Norte Glacier 2008/2009 (local LR). The lapse rate configurations 1–3 are described in Table IV. TOPKAPI-ETH hourly outputs are averaged to mean daily values. NSE, Nash and Sutcliffe efficiency;

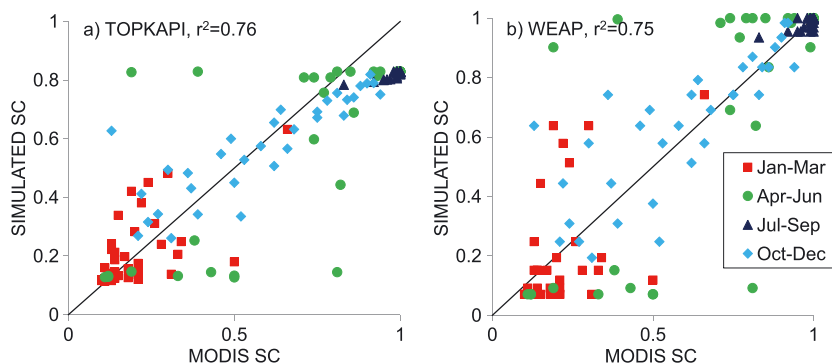


Figure 7. Moderate-resolution Imaging Spectroradiometer (MODIS) observed snow cover (SC) and snow cover simulated by the models TOPKAPI-ETH (a) and WEAP (b), Juncal River Basin, validation period (01/04/2007–31/03/2010), lapse rate configuration 1 (Table IV). r^2 is the coefficient of determination between observed and simulated snow cover

Figure 2). See also Section on Value of Local Information for a discussion of this point.

Sensitivity to a temperature increase

Figure 8 shows differences in runoff and total melt simulated by both models, considering the projections for future air temperature changes. Five points for each season represent the relative changes calculated for 5 years' simulation time (from 01/04/2005 to 31/03/2010).

The two-dimensional projection by NMDS results in a clustering of points representing April–June and July–September model responses, whereas October–December and January–March results are more scattered. Proximity of points in the NMDS graphs indicates similarity in model reaction (axes are dimensionless); therefore, Figure 8 indicates that the catchment summer response to a warmer climate varies from year to year, depending on the snow conditions. Smooth response surfaces are fitted over the limits of the NMDS graphs and provide ranges of simulated model reactions. In each subplot, point sizes are proportional to the absolute value of the variable being represented. As the location of points representing modelling results of 3-month periods is identical in each subplot, model reaction can be directly compared among models and variables. Triangles indicate positive relative changes, and inverted triangles negative relative changes.

Differences between the two models are considerable, as the models show opposite trends in late-summer (Jan–March) simulated flows: Whereas TOPKAPI-ETH predicts a general increase of late-summer runoff with respect to the present, WEAP predicts a strong decrease for most of the years (Figure 8a). These differences can be explained by the differences in simulated total melt (Figure 8b). Melt onset in WEAP is shifted to an earlier period, with less total melt sometimes already in early summer (Oct–Dec). In contrast to this, total melt in TOPKAPI-ETH increases almost throughout the year and especially for late summer (Figure 8b). The increase in

Table VI. Mean runoff and snow cover (SC) for TOPKAPI-ETH and WEAP when run with the mean gradients (configuration 1 of Table IV) and difference (%) when the models are run with precipitation and temperature lapse rates corresponding to the upper and lower bounds of the observed annual lapse rates (Figure 3)

		TOPKAPI-ETH		WEAP	
		Average	Difference (difference in %)	Average	Difference (difference in %)
Average gradient	Q	6.38 m ³ s ⁻¹		5.88 m ³ s ⁻¹	
	SC	0.5261		0.5895	
Uncertainty in precipitation gradients					
Lower bound	Q	5.08 m ³ s ⁻¹	-1.31 m ³ s ⁻¹ (-20.46)	4.17 m ³ s ⁻¹	-1.72 m ³ s ⁻¹ (-29.17)
	SC	0.5011	-0.0250 (-4.76)	0.5637	-0.0258 (-4.38)
Upper bound	Q	7.79 m ³ s ⁻¹	1.41 m ³ s ⁻¹ (22.10)	7.65 m ³ s ⁻¹	1.76 m ³ s ⁻¹ (29.97)
	SC	0.5487	0.0226 (4.29)	0.6140	0.0245 (4.16)
Uncertainty in temperature lapse rates					
Lower bound	Q	6.70 m ³ s ⁻¹	0.31 m ³ s ⁻¹ (4.92)	5.97 m ³ s ⁻¹	0.08 m ³ s ⁻¹ (1.43)
	SC	0.5185	-0.0076 (-1.44)	0.5805	-0.0089 (-1.51)
Upper bound	Q	6.11 m ³ s ⁻¹	-0.27 m ³ s ⁻¹ (-4.29)	5.80 m ³ s ⁻¹	-0.08 m ³ s ⁻¹ (-1.34)
	SC	0.5332	0.0071 (1.35)	0.5980	0.0086 (1.46)

WEAP total melt for the July–September period leads to an increase especially in October–December simulated flows.

DISCUSSION

Model performance

Considering the variability in precipitation gradients and temperature lapse rates together with the resulting variations in model outputs (Table VI), it is not surprising that model efficiencies with respect to runoff improve if annual lapse rates are applied (Table VII). The occurrence of ENSO phenomena might cause the strong interannual variability of precipitation distribution: The year 2008 – for which the shallowest precipitation gradients were observed (Figure 3) – is in fact a strong La Niña year

according to the MEI (Section on Hydrometeorological Historical Records and DEM). The year 2007 – for which we have calculated the steepest lapse rates – is the year with the most positive MEI values during the study period. Our results confirm that this variability should be taken into account independently from the type of model.

On the other hand, Table VII shows that model efficiencies with respect to MODIS SC change only little for different lapse rates and simulation periods. It is important to stress that snow accumulation and ablation occur for the most part during distinct, non-overlapping seasons: Summers are dry and characterized by a gradual decrease in SC (Oct–Dec values in Figures 5 and 7), whereas short winter precipitation events lead to complete basin snow coverage within a very short period (Apr–Jun values in Figures 7 and 5). Under the present climate,

Table VII. TOPKAPI-ETH (T) and WEAP (W) model efficiencies for the calibration (01/04/2005–31/03/2007) and validation (01/04/2007–31/03/2010) periods

	Calibration period		Validation period				
	T	W	T (1)	W (1)	T (2)	W (2)	T (3)
Runoff							
NSE	0.90	0.87	0.52	0.17	0.69	0.77	0.88
ΔQ (m ³ s ⁻¹)	0.19	0.47	0.82	0.43	0.63	0.19	-0.44
MODIS snow cover							
r^2	0.79	0.77	0.76	0.75	0.76	0.75	0.70
RMSE (%)	13.59	14.37	16.92	17.50	16.98	17.50	18.30
Δ Max (%)	-17.10	0.00	-17.10	0.00	-17.10	0.00	-17.10
Δ Min (%)	-0.60	-5.80	1.40	-2.90	0.90	-4.50	2.80

Lapse rate configurations of validation steps 1, 2 and 3 are described in Table IV. NSE is the Nash and Sutcliffe efficiency, ΔQ indicate the mean difference between observed and modelled streamflow, r^2 is the coefficient of determination RMSE the root mean square error and Δ Max and Δ Min indicate the difference between MODIS maximum snow cover and maximum simulated snow cover and between minimum MODIS snow cover and minimum simulated snow cover calculated over the entire simulation period.

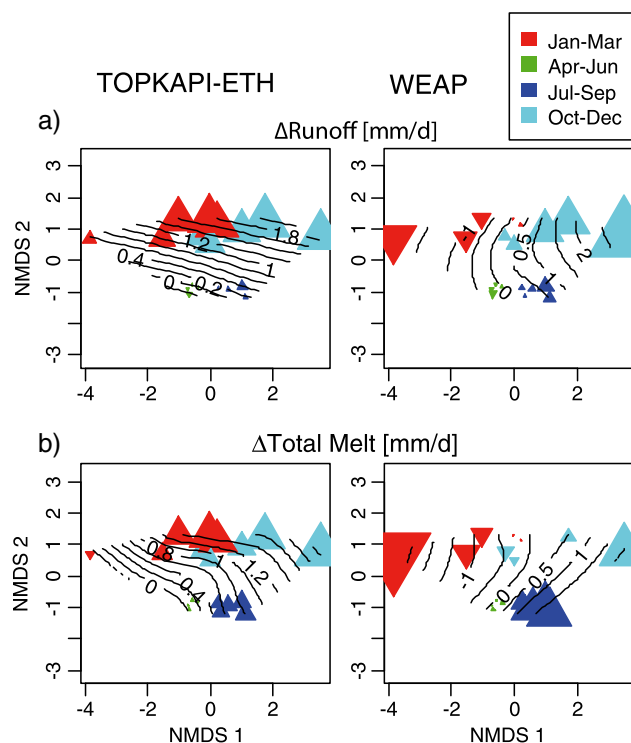


Figure 8. TOPKAPI-ETH and WEAP relative changes in modelled streamflow (a) and total melt (b), for the future temperature scenario described in Table V. A non-metric multi-dimensional scaling (NMDS) approach is used to map point objects in a two-dimensional reduced space. In the space of the two NMDS axes, similarities among model results for each 3-month period are preserved, which allows fitting the smooth response surfaces over the plots. Point-up triangles indicate positive relative changes, and point-down triangles indicate negative relative changes in simulated variables. The size of triangles is proportional to the absolute value of modelled relative changes

precipitation gradients are the firstmost control on snow accumulation at the mesoscale. Overall goodness of fit with respect to SC (Table VII) will therefore always be more affected by the much longer period of SC decrease and are not very sensitive to the variability in precipitation gradients (Table VI).

As a consequence, model calibration in WEAP with MODIS binary product is likely affected by snow melt, rather than snow accumulation processes. Accordingly, WEAP often overestimates SC after summer and autumn storm events during calibration (SC of 0.9 or higher in Figure 5b, whereas MODIS SC is lower than 0.6), although the decrease in SC is reproduced well (Figure 5b). On the other hand, the small changes in SC goodness-of-fit measures for different simulation periods (Table VII) demonstrate that for both WEAP and TOPKAPI-ETH, it is suitable to use constant melt parameters over time. Melt parameter stability is likely favoured by the stable meteorological conditions that are typical of the ablation seasons in the Central Andes of Chile (Ragetti and Pellicciotti, 2012).

Value of local information

Table VII shows that TOPKAPI-ETH performance with respect to observed runoff increases if increasingly better data are available for parameter estimation (such as the higher temporal and spatial resolution data sets for the lapse rate configurations, Table IV). The results of the validation steps indicate that up to some degree, the accuracy of model output can be directly related to the accuracy in model input also outside the calibration period. The same is true for WEAP, except that the semi-distributed spatial representation by elevation bands does not allow taking into account different temperature distributions for different surface types. As such, WEAP is limited in its capacities to incorporate local information about atmosphere–surface interactions in order to enhance model performance and cannot make efficient use of the locally collected observations.

Process representation

Both models are able to simulate runoff adequately for the present climate (Table VII) but predict a very different response to a climate change scenario (Figure 8). In order to explain this difference, we first look at the water balance calculated by each model with annual meteorological lapse rates, which lead to an adequate fit with respect to measured runoff for both models (Table VII). Mean annual values of the water balance components are shown in Table VIII for the calibration and validation periods, as well as for both periods combined. The most important difference concerns the input of glacier melt, which is a model component not considered by WEAP. Glacier contribution to streamflow estimated by TOPKAPI-ETH is on average 20% over the entire year (Table VIII) and is larger than the glacierized watershed fraction (14%). The lower melt in WEAP is compensated by lower evapotranspiration and less annual net snow accumulation (snow remnant), which in turn leads to similar results regarding mean annual basin outflow (runoff). Mean annual snow remnants calculated for the entire period are higher in TOPKAPI-ETH than in WEAP, but still smaller than the glacier ice-melt input, which indicates negative glacier mass balance. This is in agreement with recent observations of glacier retreat in the area by Bown *et al.* (2008). Our results indicate that even for a modest percentage of glacierization, predicting hydrologic responses to climate change requires that glacier hydrology be taken into account, despite the uncertainty that might affect the associated parameters in empirical models.

Total inputs in the form of snowfall and rain are about the same for TOPKAPI-ETH and WEAP, although the partitioning differs slightly: In WEAP, the liquid precipitation input is slightly smaller than that in TOPKAPI-ETH, especially during the calibration period, which supports our conclusions that WEAP might overestimate late-summer and autumn snow accumulation

as discussed in the Section on Model Performance. Total precipitation inputs are more than 30% higher for the calibration period than for the validation period (Table VIII). The differences in total precipitation between the two periods allow an analysis of the ability of the models to adapt to different snow situations. The simplifying assumption made in WEAP of ignoring glacier processes becomes less valid, as meltwater inputs from glaciers increase when they are less covered by snow. Accordingly, glacier contribution simulated by TOPKAPI-ETH increases by about 40% with respect to the calibration period (Table VIII). Underestimation of simulated late-summer flows by WEAP can be observed especially for 2009/2010 (Figure 6), which was a very dry year at high elevations. For the year 2008/2009, characterized by lower-than-usual precipitation at high elevations (Figure 3) and where glacier contribution to total streamflow was important (Ragettli and Pellicciotti, 2012), WEAP does not underestimate late-summer runoff (Figure 6). However, because maximum runoff is overestimated earlier the same year, it is likely that WEAP overestimates total snow meltwater inputs, thus compensating for the missing glacier contribution.

The PCA biplots in Figure 9 provide more insights about the differences in process representation reported in Table VIII. Arrows characterize descriptor axes for each variable considered, and points represent model results for 3-month periods. Because of the strong seasonality to which almost all hydrological processes in the Central Andes of Chile are subject, the PCA plots can be read in a clockwise sense, where one complete turn represents the full sequence of dominant processes within one hydrological year.

For TOPKAPI-ETH, the beginning of the hydrological year is characterized by precipitation in the form of both snow and rain (Apr–Jun; Figure 9a). Late winter (Jul–Sept) is still marked by heavy snowfall, whereas October–December correlates with snowmelt at lower

elevations. Late summer (Jan–March) is characterized by snowmelt at high elevations and ice melt (Figure 9a). This sequence is about the same in the case of WEAP, with the difference that the melting season starts earlier – already between July and September (Figure 9b). Accordingly, October–December flows correlate more with snowmelt between 4000 and 5000 m a.s.l.. A clear difference between models concerns the soil moisture vector, which points more towards late summer in WEAP and towards spring–summer in TOPKAPI-ETH. This indicates that meltwater is stored in the soil for longer in the season in WEAP. However, soil moisture reaching its maxima in early summer is more realistic, considering that the main snowmelt season in the parts of the valley where soils are located (below 4000 m a.s.l.) is October–December. The overestimation of baseflow in winter by WEAP (Figures 4 and 6) is another indication that soil storage capacities are overestimated. Figure 10 confirms that annual changes in soil water storage in WEAP are more important than those in TOPKAPI-ETH. The net change in soil moisture storage during the main melting season (Oct–Dec) is more than 100 mm higher in WEAP, which is 60% of the mean annual glacier ice melt simulated by TOPKAPI-ETH (Table VIII).

Figure 11 depicts the temporal and spatial distributions of snowmelt and snow remnants within the Juncal River Basin and supports the findings about the earlier melt onset in WEAP with respect to TOPKAPI-ETH. The main period of snowmelt is October–December, when according to both models, the largest fraction of snow is melted (Figure 11a). Above 4000 m a.s.l., a considerable fraction of snow melts from January to March. Below 3000 m a.s.l., according to TOPKAPI-ETH, most of the snow melts between July and September. This sequence is visibly shifted in the case of WEAP, with more snow being melted earlier in the season in each elevation band and less net snow accumulation at the end of the season

Table VIII. Mean annual values of the components of the water balance for the period April 2005–March 2010, using annual individual gradients for model simulations

	TOPKAPI-ETH	WEAP
Total inputs		
Precipitation, solid	1081.7 (1428.4/1049.9)	1125.8 (1370.2/962.8)
Precipitation, liquid	119.6 (125.7/115.5)	67.3 (48.8/79.6)
Glacier ice melt	180.8 (145.8/204.2)	—
Residuals		
Snow remnants	129.3 (183.4/93.4)	31.2 (56.3/14.4)
Soil water volume	−47.3 (−19.2/−66.0)	−6.7 (8.6/−17.0)
Losses		
Evapotranspiration	−413.9 (411.5/415.5)	−304.2 (332.9/285.0)
Outflow	−883.7 (995.9/809.2)	−864.5 (1021.2/760.0)

In brackets are the mean annual values for the calibration/validation periods, respectively. Values are expressed in millimetre water equivalents per year.

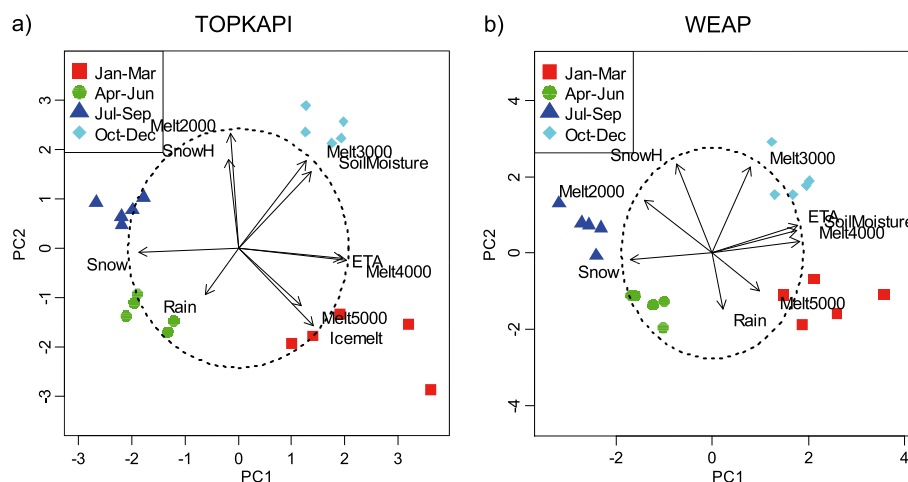


Figure 9. TOPKAPI-ETH and WEAP correlation biplot of the principal component (PC) analysis: double projection of point objects (3-month periods) and descriptor axes (variables) on a reduced space. The cosine of an angle between two descriptor axes represents the correlation between the two variables. Orthogonal projection of point objects on descriptor axis indicates the relative position of 3-month periods along variables. The variables that are considered are precipitation in liquid form (rain), precipitation in solid form (snow), basin-wide average snow depth (SnowH), snowmelt for different elevation zones [<3000 m above sea level (a.s.l.): Melt2000, 3000–4000 m a.s.l.: Melt3000, 4000–5000 m a.s.l.: Melt4000, and >5000 m a.s.l.: Melt5000], basin-wide average soil water content (SoilMoisture) and actual evapotranspiration (ETA). Values are average of the simulations over 5 years (01/04/2005–31/03/2010) with lapse rate configuration 2 (described in Table IV)

(Figure 11b). The earlier melt onset simulated by WEAP compensates for the overestimation of precipitation in the form of snow (Section on Model Performance, Table VII): A deeper snowpack means that more snow needs to be melted before SC can begin to decrease, in order to match the MODIS SC. Because the inputs to the hydrological system are overestimated, overestimation of soil storage capacity in the second calibration step occurs, in order to hold the water in the soil for a longer period before it is routed to the outlet. This is a classical equifinality problem, where errors in components of the the basin hydrology compensate each other and as a result more than one combination of parameters provides a good agreement with runoff (e.g. Beven, 2001, 2002). This might not be relevant for the present climate but has important consequences when simulating future warmer conditions. It explains why runoff simulated by WEAP for the climate scenario for the period October–December increases with respect to the present climate, whereas total melt computed for the same period often decreases (Figure 8).

Snow accumulation. In order to explain the differences in net snow accumulation between the two models (Table VIII and Figure 11), we computed the distribution of March snow remnants in space (Figure 12a) and averaged it over elevation bands. Differences in net snow accumulation between models occur between 3500 and 4500 m a.s.l. (Figure 12a). Above 4500 m a.s.l., both models show net snow accumulation. According to Carrasco *et al.* (2005), the equilibrium line altitude (ELA, which corresponds to the altitude where the annually integrated mass balance is zero) at 33°S in the Central Andes of Chile is approximately at

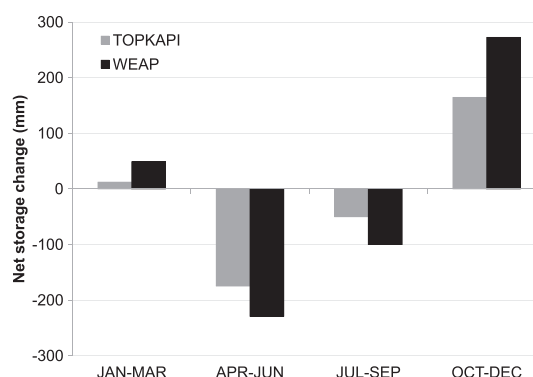


Figure 10. Seasonal soil storage changes in WEAP and TOPKAPI-ETH. Negative storage change indicates a decrease of soil water volume, positive storage changes an input to soil water storage. Values represent the mean net storage change over 3-month periods, averaged from the simulations over 5 years (01/04/2005–31/03/2010) with lapse rate configuration 2 (described in Table IV)

4200 m a.s.l. The ELA rises from the west to the east in the Central Andes (Lliboutry, 1998). Therefore, using the ELA as a possible indication of the transient snowline, the end-of-summer snowline at 4500 m a.s.l. simulated by WEAP seems plausible. However, the more irregular line of TOPKAPI-ETH in Figure 12a reflects better the topographically driven snow variability within the basin. Snow remnants at high elevations depend on topographical features such as aspect and shading and on snow accumulation due to avalanches, which cannot be represented in WEAP because of its spatial structure and lack of a snow redistribution routine. The end-of-summer SC distribution between 3000 and 4500 m a.s.l. simulated by TOPKAPI-ETH agrees very well with MODIS observed end-of-summer

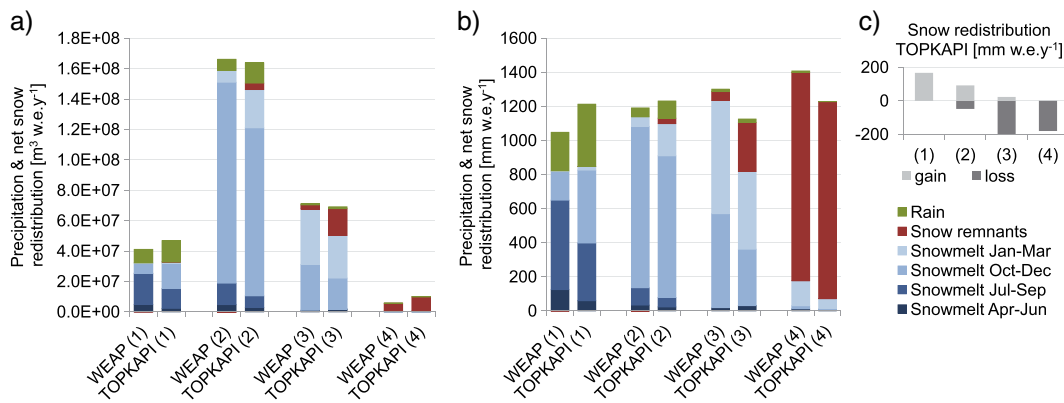


Figure 11. Mean annual snow remnants and 3-month seasonal snowmelt per 1000-m elevation band: (a) values in cubic metre water equivalents per year, (b) values in millimetre water equivalents per year, (c) amount of snow (mm w.e. year^{-1}) that is redistributed from one elevation band to another in TOPKAPI-ETH. Elevation bands: (1) <3000 m, (2) 3000–4000 m, (3) 4000–5000 m and (4) >5000 m. Values indicated are the average of simulations over 5 years (01/04/2005–31/03/2010) with lapse rate configuration 2 (described in Table IV).

SC (Figure 12b) and confirms that topographically induced snow redistribution plays an important role in the area: A semi-distributed model cannot reproduce the scattered SC distribution in regions of complex terrain, and as a consequence, the simulated WEAP SC is unrealistically associated to a given elevation threshold (Figure 12b).

Above 5000 m a.s.l., TOPKAPI-ETH-simulated end-of-summer SC is lower than the MODIS one. To assess how realistic this representation of SC is, we compare the results with the SC fraction obtained from end-of-summer SC data derived from Landsat 5. Figure 12b shows that TOPKAPI-ETH results and Landsat data agree well at high altitudes, which confirms that the MODIS spatial resolution (500 m) does not fully reflect the complex snow patterns in regions of very steep topography. Between 3000 and 4000 m a.s.l., TOPKAPI-ETH-simulated SC agree well with both Landsat and MODIS end-of-summer remnants. Between 4000 and 5000 m a.s.l., Landsat data indicate less SC than TOPKAPI-ETH and MODIS. This altitudinal range is characterized by very

steep slopes (Figure 13), and it might be possible that also the TOPKAPI-ETH 100-m resolution is not sufficient to reproduce snow patterns more accurately. However, agreement is again very good above 5000 m, indicating that the reduced SC at very high elevation simulated by TOPKAPI-ETH (Figure 12a) is true and reflects the movement of snow from the high, steep upper reaches to the lower ones.

Because WEAP cannot reproduce the aggregated distribution of snow at high elevations (Figure 12b), this affects the calibration of melt parameters against MODIS data and might explain why WEAP underestimates summer SC. Figure 13 provides evidence about the effect of the missing topographical control on the calibrated WEAP SC: During the main melting season (Oct–Dec, Figure 11), the agreement of simulated with observed SC is on average up to 10% lower for regions with steep slopes between 3000 and 4000 m a.s.l. The altitudinal range between 3000 and 4000 m a.s.l. represents more than 50% of the total area of the watershed, and 15% of

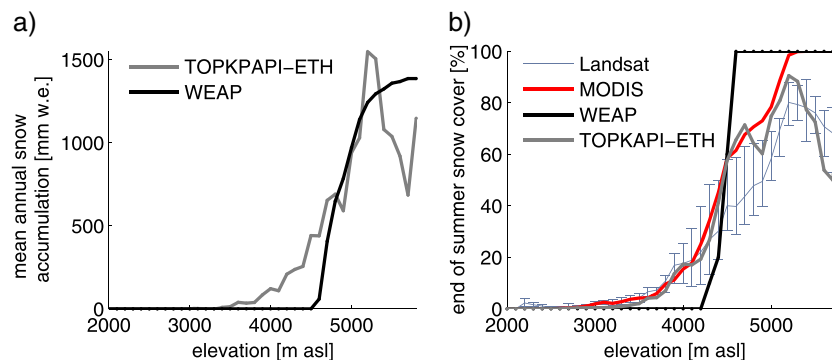


Figure 12. End-of-summer snow accumulation and snow cover per elevation band: (a) mean annual net snow accumulation simulated by TOPKAPI-ETH and WEAP per 100-m elevation band, (b) comparison of simulated (TOPKAPI-ETH and WEAP) and observed [MODIS and Landsat5] fractions of snow-covered area of each 100-m elevation band. Values in both panels are mean values averaged over 5 years (01/04/2005–31/03/2010) and obtained with the lapse rate configuration 2 (described in Table IV). The TOPKAPI-ETH, WEAP and MODIS values are the average of 8-day maximum snow cover of all MOD10A2 8-day periods in March with less than 20% snow cover. The Landsat line and error bars show the mean, maximum and minimum snow cover conditions obtained from four end-of-summer Landsat images (from 9 Apr 2006, 11 Mar 2007, 3 Mar 2008 and 16 Mar 2009)

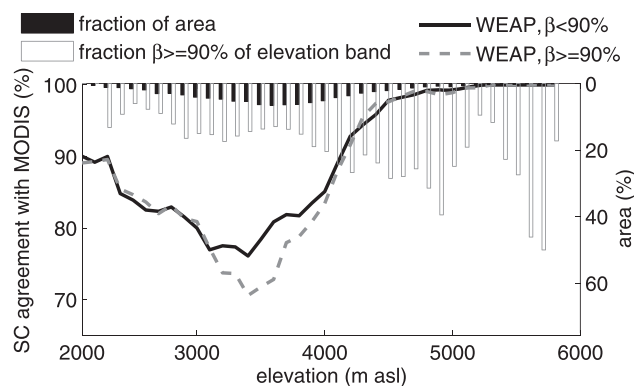


Figure 13. Mean agreement of WEAP simulated with MODIS observed snow cover (SC) during the main melting season (Oct–Dec). Both the MODIS snow cover 500-m grid maps and the WEAP 100-m elevation band snow cover maps are resampled according to the TOPKAPI-ETH 100-m grid, and the agreement is calculated separately for steep ($\beta \geq 90\%$ inclination) and non-steep cells ($\beta < 90\%$ inclination). The secondary axis shows the fraction of the total area represented by each elevation band and the fraction of area of each elevation band with steep cells. Snow cover is obtained as the mean value of simulations over 5 years (01/04/2005–31/03/2010) with lapse rate configuration 2 (described in Table IV)

this elevation zone has an inclination greater than or equal to 90% (Figure 13). Eighty-three per cent of the snow from this altitudinal range is melted between October and December, according to WEAP simulations (Figure 11). A semi-distributed model that does not redistribute snow from steep slopes regards the absence of snow as snow that has been melted, and not as snow that has been redistributed. This leads to enhanced simulated snowmelt over a given elevation band in order to maximize the agreement between simulated and observed SC also for regions with steep slopes. As a consequence, the combined effect of slopes, aspect and shading on the aggregated distribution of snow partially reflected by the MODIS images but not considered by WEAP leads to the underestimation of simulated minimum summer SC by 3–6% by WEAP (reported in Table VII and Figure 5) and an underestimation of simulated snow remnants (Figure 12). This overestimation of melt at high elevations by WEAP explains in part why the model is able to reproduce late-summer streamflow without considering the contribution of glaciers, at least in wet years such as 2005/2006, even when error compensation is minimized by calibrating melt parameters against observations of SC and not against runoff. The difference in simulated snow remnants between the two models is equal to 90% of total simulated TOPKAPI-ETH glacier melt during the calibration period (Table VIII).

Model response to climate warming

Simulations under a temperature increase scenario have offered the chance to evaluate fully the potential and limitation of a modelling approach such as the one

adopted in WEAP that does not include glaciers explicitly. Differences in process representation between the two models are evident already for present simulations (Section on Process representation but not fully visible in the runoff simulated for the present climate thanks to error compensation. They become evident, however, and of great significance, in a warmer climate, for which the two models' responses diverge, with a significant decrease of snowmelt at the end of the melting season and strong decrease of late summer runoff for WEAP and an increase of runoff for TOPKAPI-ETH (Figure 8). It is obvious that the simulations of TOPKAPI-ETH in this case are more reliable than those of WEAP. These results show that internal consistency is crucial for model transferability, i.e. when models are applied for conditions that differ from those for which they have been calibrated, and suggest that rigorous tests of internal consistency should always be carried out in any evaluation of model performance (Schaeffli and Huss, 2011).

CONCLUSIONS

The work presented here contributes to strengthening the concept that modelling strategies based on considerations of data availability, level of knowledge of hydrological processes and perceived effort required for model construction affect predictions in ways relevant for decision-making in water resources and must be taken into account carefully in modelling studies. This effect is demonstrated for a high-elevation basin in the extratropical Andes of Central Chile by comparing the hydrologic response of a physically based, distributed and research-oriented model (TOPKAPI-ETH) and a semi-distributed conceptual model widely used in climate change hydrologic impact assessments (WEAP) under a hypothetical warmer climate. This research is not an exhaustive model intercomparison exercise, but an investigation into modelling strategies appropriate for the estimation of hydrologic impacts of climate change in poorly monitored regions.

Hydrologic predictions at temporal scales relevant for water resource management (trimestral river flows) differ significantly between models, although model skill against daily streamflow for calibration and validation is similar and quite high. The largest differences occur for late-summer (Jan–Mar) flows, i.e. during the most critical time of the year for agriculture and domestic demands, and can be traced back to differences in the representation of the chain of storage compartments in the basin, namely the seasonal snowpack, a moderate glacier cover and subsurface storage. Daily streamflow values and 8-day SC MODIS data are used for model calibration and validation. Although both models reproduce these observations adequately, the simplified structure adopted in WEAP (elevation band spatial discretization and lack

of glacier component) can lead to unexpected error compensation through the calibration process: Melt begins early in WEAP, but soil water storage variations are higher than in TOPKAPI-ETH, which buffer the earlier melt onset on the runoff simulations. Likewise, WEAP underestimates systematically late-summer snow remnants throughout the basin despite being calibrated against MODIS imagery; this can be explained by this model's lack of representation of topographically driven snow redistribution during the accumulation season. Accounting for topographically induced snow transport modifies the seasonal distribution of modelled snowmelt amounts (Warscher *et al.*, 2013), which allows models to reproduce correctly internal processes, leading to runoff generation. By showing that MODIS SC cannot be used to calibrate a semi-distributed model based on elevation bands in rough mountainous terrain in the Andes, we point to an important finding, as MODIS SC is frequently used for hydrological model calibration or validation (e.g. Parajka and Blöschl, 2008; Franz and Karsten, 2013).

Together with snow redistribution, we show that interannual variability in the spatial distribution of snowfall is a major driver of model performance. Altitudinal precipitation gradients inferred from rain gauges and snow pillow data show important year-to-year variation, and both models perform poorly in the validation period when forced with constant gradients averaged over several years of record. Along the entire Andes Cordillera, ENSO plays an important role in modulating precipitation volumes and might also play a role in determining patterns of spatial variability. Further research is needed to quantify this influence and to develop diagnostic models that help predict not only the volume but also the spatial distribution of seasonal precipitation. Detailed information on other meteorological forcings such as temperature lapse rates, which were obtained through short-term, especially designed field experiments, was successfully incorporated into the physically based distributed model. This in turn enhances the representation of all relevant processes and minimizes the risk of error compensation, with its undesirable impacts on model reliability.

In summary, our work highlights the challenges and possible strategies for dealing with hydrologic predictions in sparsely monitored mountain watersheds, underscoring those sources of information most relevant for enhancing the reliability of estimates under future climate. One of the most encouraging features of our work is the evidence that carefully planned short-term data acquisition field experiments can successfully complement long-term historical records when ingested into a physically based distributed model. Other model intercomparison studies have argued that the added complexity associated with these models

does not offset the gains in model performance compared with simpler models (e.g. Refsgaard and Knudsen, 1996) or that there is no clear link between model complexity and performance (e.g. Essery *et al.*, 2013). However, when tested under climatic conditions different from those prevalent at the time of calibration, our results highlight the need for deeper knowledge about the dominant processes in high-elevation basins in this region and their most appropriate model representation. Future work should include analogue climate modelling studies in which models are tested in paired basins for which climate differences are significant.

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